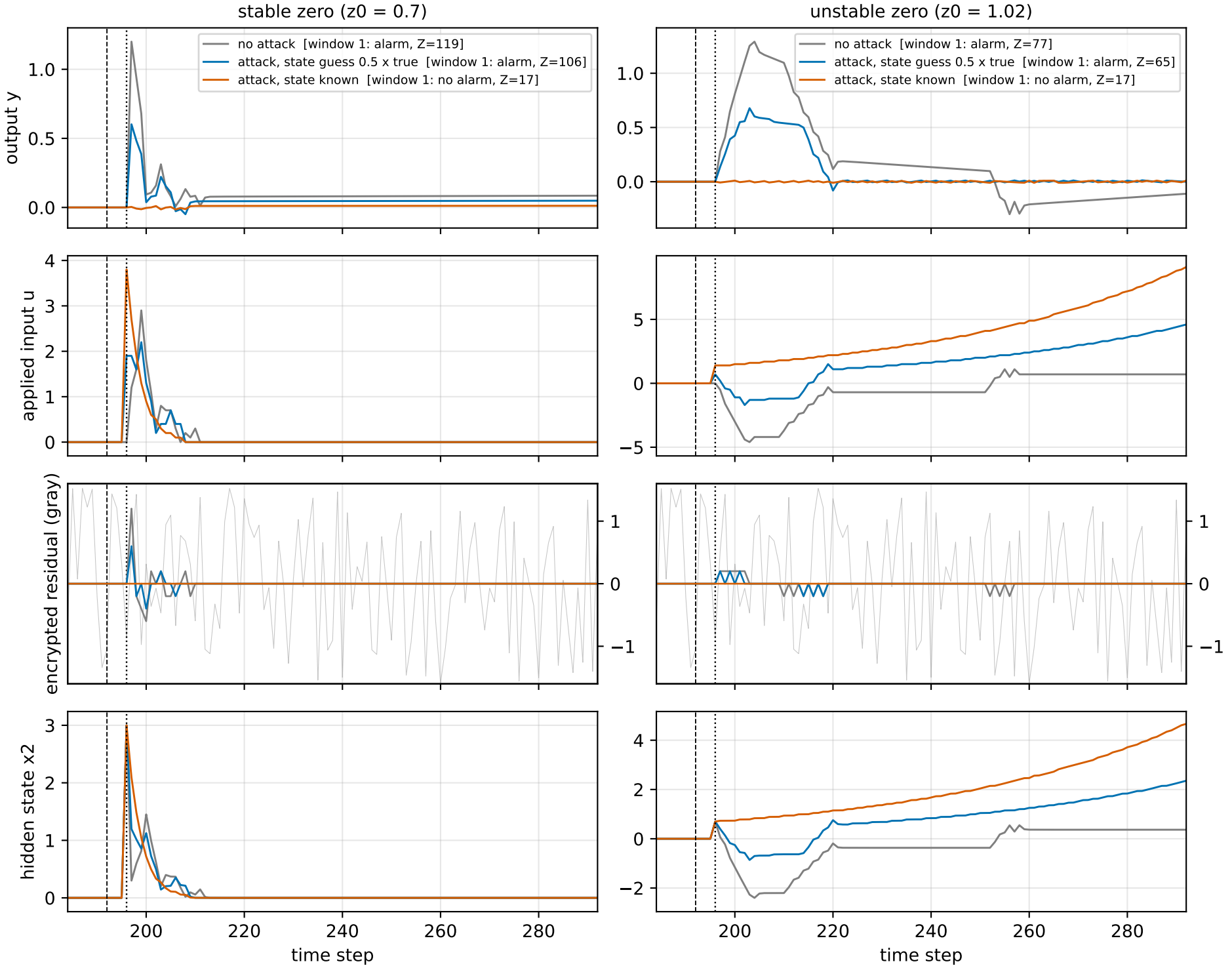
Zero-dynamics attack starting with a disturbance 4 steps into window 1 (dotted); encrypted loop
stable zero (z0 = 0.7)
unstable zero (z0 = 1.02)
no attack [window 1: alarm, Z=119]
attack, state guess 0.5 x true [window 1: alarm, Z=106]
attack, state known [window 1: no alarm, Z=17]
no attack [window 1: alarm, Z=77]
attack, state guess 0.5 x true [window 1: alarm, Z=65]
attack, state known [window 1: no alarm, Z=17]
output y
applied input u
encrypted residual (gray)
hidden state x2
time step
time step